



Pizza Sales Analysis using SQL

Project Overview

- Pizza Hut is one of the world's leading pizza restaurant chains.
- The objective of this project is to analyse pizza sales data using SQL.
- By analysing customer orders, pizza categories, sizes, and revenue, we can identify business trends and opportunities.

SQL | Data Analysis | Business Insights



The project demonstrates SQL concepts such as:

- ❑ Joins
- ❑ Aggregate
- ❑ Functions
- ❑ GROUP BY
- ❑ Subqueries
- ❑ Window Functions

Tools Used:

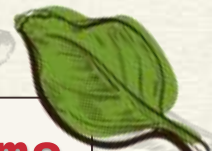
Excel/CSV Dataset
MySQL
PowerPoint



Project Objectives

- Analyse sales performance.
- Customer Preferences.
- Identify top and low-performing products.
- Calculate revenue contributions.
- Support data-driven business decisions.





Dataset Overview & Database Schema

Dataset Information

This project contains 4 relational tables

Table	Description
Orders	Stores order date and time information
Order_Details	Stores pizza orders and quantities
Pizzas	Stores pizza sizes and prices
Pizza_Types	Stores pizza names and categories





Dataset Overview & Database Schema

Table Relationships

Orders

|

| Order_ID

|

Order_Details

|

| Pizza_ID

|

Pizzas

|

Pizza_Type_ID

|

Pizza_Types



Dataset Fields



1. Orders

- ❖ Order_ID
- ❖ Date
- ❖ Time



2. Order_Details

- ❖ Order_Details_ID
- ❖ Order_ID
- ❖ Pizza_ID
- ❖ Quantity



3. Pizzas

- ❖ Pizza_ID
- ❖ Pizza_Type_ID
- ❖ Size
- ❖ Price



4. Pizza_types

- ❖ Pizza_Type_ID
- ❖ Name
- ❖ Category



Retrieve the total number of orders placed.

```
SELECT  
    COUNT(order_id) AS Total_orders  
FROM  
    orders;
```

Insight:

- ❖ Total customer orders placed during the analysis period = **21,350**.

	Total_orders
▶	21350





Calculate the total revenue generated from pizza sales.

```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),2) AS Total_Revenue
FROM
    order_details
    INNER JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Insight:

- ❑ Total Revenue Generated = **\$817,860.05**
- ❑ Revenue is calculated by multiplying the quantity sold by the pizza price.

	Total_Revenue
▶	817860.05



Identify the highest-priced pizza. (Top 2)

```
SELECT
    pizza_types.name, Pizzas.price
FROM
    pizza_types
    INNER JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 2;
```

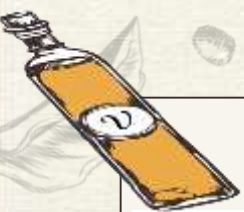


Insight

- ❑ The Greek Pizza is the highest-priced pizza.
- ❑ Premium pizzas can generate higher revenue per sale.

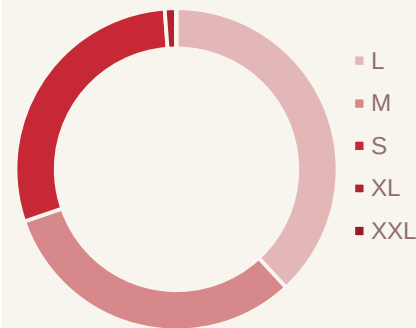
	name	price
▶	The Greek Pizza	35.95
	The Greek Pizza	25.5





Identify the most common pizza size ordered.

```
SELECT
  pizzas.size,
  COUNT(order_details.order_details_id) AS Total_Quantity
FROM
  pizzas
  INNER JOIN
  order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizzas.size
ORDER BY Total_Quantity DESC;
```



Insight:

- ❑ Large (L) pizzas are the most frequently ordered size.
- ❑ Medium and Small sizes are also popular.
- ❑ XL and XXL pizzas have relatively low demand.
- ❑ Pizza Hut can focus promotions and inventory planning on larger sizes
- ❑ Customers show a strong preference for **regular-sized pizzas (L, M, S)**.
- ❑ Promotions and inventory planning should primarily focus on **Large-sized pizzas..**

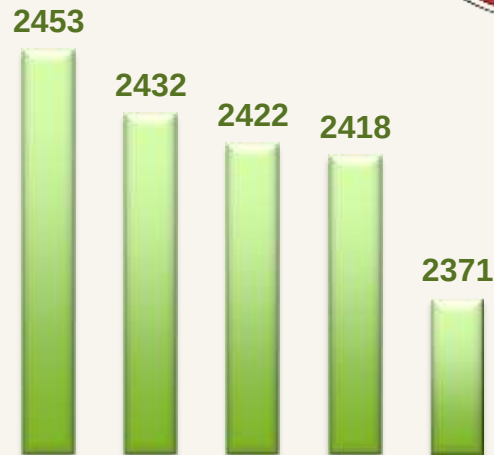
	size	Total_Quantity
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28





List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity) AS Total_quantity
FROM
    Pizza_types
    INNER JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    INNER JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Total_quantity DESC
LIMIT 5;
```



	name	Total_quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Insight:

- ❑ The Classic Deluxe Pizza is the highest-selling pizza, with **2,453** units sold, indicating strong customer preference for classic flavours.



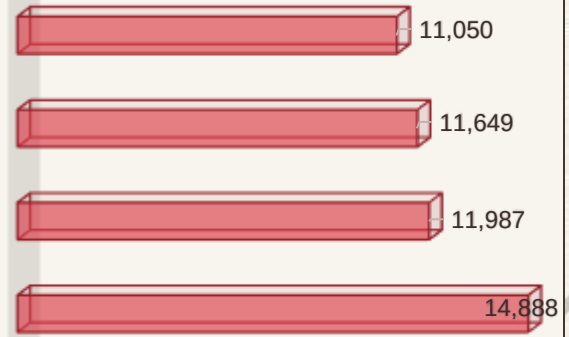


Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS Total_Quantity
FROM
    pizza_types
    INNER JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    INNER JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY Total_Quantity DESC;
```

Insight:

- ❑ Classic pizzas are the highest-selling category, accounting for **14,888** units ordered.
- ❑ Chicken pizzas have the lowest order quantity among all categories.
- ❑ Demand is relatively balanced across categories, with no category showing an extreme gap in sales.
- ❑ The strong performance of Classic pizzas indicates consistent customer preference for traditional flavours.



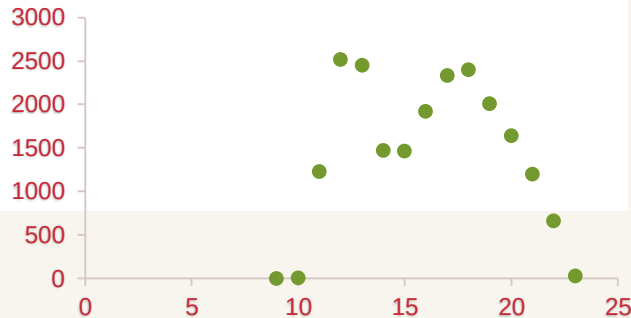
	category	Total_Quantity
►	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050





Determine the distribution of orders by hour of the day.

```
SELECT  
    HOUR(time) AS Order_hour, COUNT(order_id) AS Total_orders  
FROM  
    orders  
GROUP BY HOUR(time)  
ORDER BY Order_hour;
```



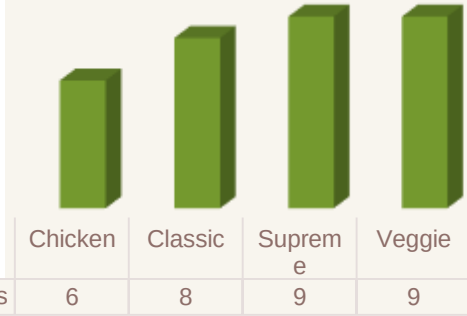
Insights:

- ✓ Peak ordering time is **12 PM**, recording **2,520** orders.
- ✓ **1 PM** is the second busiest hour with **2,455** orders, indicating a strong lunch rush.
- ✓ A second demand peak occurs during **5 PM–7 PM**, reflecting dinner-time customer activity.
- ✓ Order volume drops significantly before **11 AM** and after **10 PM**.
- ✓ Pizza Hut can optimise staffing and inventory during **12 PM–1 PM** and **5 PM–7 PM** to meet peak demand.

	Order_hour	Total_orders
▶	9	1
	10	8
	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28

Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name) AS Total_pizza
FROM
    pizza_types
GROUP BY category
ORDER BY Total_pizza;
```



Insight:

- Supreme and Veggie categories offer the widest variety, with **9** pizza types each.
- The Classic category contains **8** pizza types.
- The Chicken category has the fewest options, with **6** pizza types.
- The menu is fairly balanced across categories, providing customers with diverse choices.

	category	Total_pizza
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 2) AS Average_Pizza_Per_Day
FROM
    (SELECT
        orders.date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    INNER JOIN order_details ON order_details.order_id = orders.order_id
    GROUP BY orders.date) AS Daily_Orders;
```

Insight:

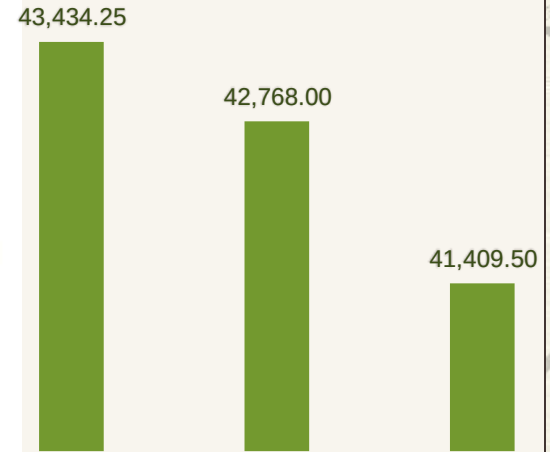
- On average, **138.47** pizzas were ordered per day, providing a reliable benchmark for demand forecasting and operational planning.



Average_Pizza_Per_Day	
▶	138.47

Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS Revenue
FROM
    pizza_types
    INNER JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    INNER JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Revenue DESC
LIMIT 3;
```



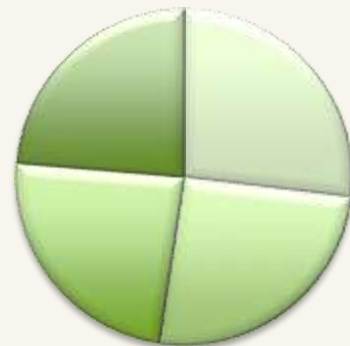
Insight:

- ❖ The top 3 pizza types generated the highest revenue, indicating strong customer preference and significant contribution to overall sales. These pizzas should be prioritised in promotions and inventory planning.

	name	Revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    pizza_types.category,
    round((SUM(order_details.quantity * pizzas.price) / (SELECT
        SUM(order_details.quantity * pizzas.price) AS Total_Sales
    FROM
        order_details
        INNER JOIN
        pizzas ON pizzas.pizza_id = order_details.pizza_id)) * 100,2) AS revenue_percentage
FROM
    pizza_types
    INNER JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    INNER JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue_percentage DESC;
```



■ Classic ■ Supreme
■ Chicken ■ Veggie

Insight :

- Classic pizzas accounted for **26.91%** of total revenue, making them the top-performing category. Veggie pizzas accounted for the lowest share (**23.67%**), indicating lower revenue generation than the other categories.

	category	revenue_percentage
►	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68



Analyse the cumulative revenue generated over time.

```
SELECT Date,  
       round(SUM(revenue) OVER (ORDER BY date),2) AS cum_revenue  
FROM (  
  SELECT orders.date,  
         SUM(order_details.quantity * pizzas.price) AS revenue  
  FROM order_details  
  INNER JOIN pizzas ON order_details.pizza_id = pizzas.pizza_id  
  INNER JOIN orders ON orders.order_id = order_details.order_id  
  GROUP BY orders.date  
) AS sales;
```

Insight:

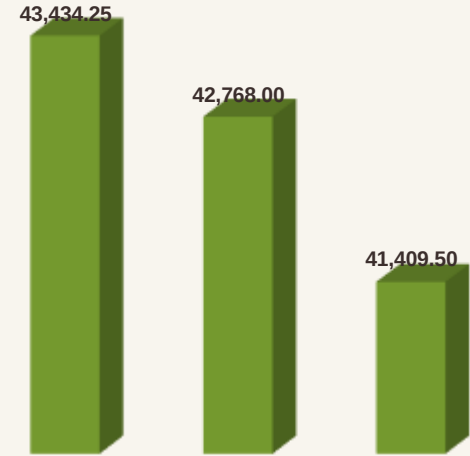
- ❑ The cumulative revenue curve maintained a steady upward trajectory without significant stagnation, indicating stable revenue generation and a positive growth trend throughout the analysis period.





Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
Select Name, revenue from
(select category, name, revenue, rank() over(partition by category order by revenue desc) as rn
from
(Select pizza_types.category, pizza_types.name,
SUM(order_details.quantity*pizzas.price) AS revenue
from pizza_types inner join pizzas
on pizza_types.pizza_type_id=pizzas.pizza_type_id
inner join order_details
on order_details.pizza_id=pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as A)as b
where rn<=3 LIMIT 3;
```



Insight:

- The top 3 pizzas in each category accounted for a significant share of category revenue. These products are the most valuable menu items and should be prioritised in promotions, inventory planning, and customer retention strategies.

	Name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

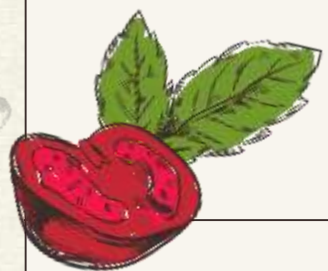




CONCLUSION

- ❖ This Pizza Sales Analysis transformed raw order data into actionable business insights by analysing customer ordering patterns, revenue trends, product performance, and sales distribution. The analysis demonstrated how SQL can be used to support data-driven decision-making in the food and restaurant industry.

Recommendations:

- 📈 Consistent Revenue Growth
 - 🍕 Top Pizzas Drive Majority of Revenue
 - 🕒 Peak Ordering Hours Identified
 - 👤 Clear Customer Preference Patterns
 - 🎯 Opportunities for Targeted Promotions
 - 💰 Data-Driven Decisions to Improve Profitability
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Cover Page

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